^{31}P(α,p),(α,n):resonances 1975Sc40,1971Mc23,1964Ku12

$J^{\pi}=1/2^{+}$ ^{31}P ground state.

1975Sc40: ^{31}P(α,p) with 3.25-5.55-MeV α beams produced from the 5.5-MV Van de Graaff accelerator of the University of Lowell. The target was prepared by evaporating natural ^{31}P onto carbon backings. Protons were detected using four silicon surface-barrier detectors. Measured proton yields and angular distributions. Deduced resonance levels at E_x=9.9-11.8 MeV, J, and π.

1971Mc23: ^{31}P(α,p) with 3.25-5.25-MeV α beams produced from the 5.5-MeV Van de Graaff accelerator of the Southern Universities Nuclear Institute. The target was Zn_3P_2 evaporated onto carbon backings. Protons were detected using four solid state detectors. Measured proton yields and angular distributions. Deduced resonance levels at E_x=9.9-11.7 MeV, J, and resonance strengths.

1964Ku12: ^{31}P(α,p) with 1.7-3.3-MeV α beams produced from the Utrecht Van de Graaff accelerator. A zinc phosphide target was prepared by evaporation of 40-500- mg/cm^2 Zn_3P_2 onto solid copper backings. Protons were detected using five silicon surface barrier counters at 87°, 120°, 135°, 150°, and 172°. Measured resonance yields and proton angular distributions. Deduced ^{35}Cl resonance energies, J, and resonance strengths.

1970Um01: ^{34}S(p,n), ^{31}P(α,n), ^{31}P(α,α_0) reactions with proton and α beams produced from the Florida State University Tandem Van de Graaff accelerator. Targets for ^{34}S(p,n) were a CdS target by evaporating ^{34}S enriched CdS onto 0.12-mm gold plate and a AgS target heating ^{34}S (67% enriched) in the presence of 0.025-mm Ag foil. Targets for ^{31}P(a,n) were prepared by evaporating natural Ca_3P_2 onto 0.075-mm Ta plate and 0.025-mm Al foil. Positrons from ^{34}Cl decay were detected using an NE102 plastic scintillator. γ rays in ^{34}S following the decay were detected using a Ge(Li) detector. Measured excitation functions. Deduced level energies and isospin T=3/2 for ^{35}Cl resonance states at 12.9 and 13.9 MeV.

^{35}Cl Levels

(2J+1)$\Gamma_\alpha\Gamma_p/\Gamma$: from 1964Ku12 for E_x<9950 and from 1971Mc23 for E_x>9950.

E(level)†	J^{π}‡	E_α(lab) (keV)	Comments
9127 *9*	5/2#	2404# *10*	(2J+1)$\Gamma_\alpha\Gamma_p/\Gamma$=0.16 eV.
9161 *4*	1/2#	2442# *4*	(2J+1)$\Gamma_\alpha\Gamma_p/\Gamma$=1.6 eV.
9256 *4*	1/2#	2550# *4*	(2J+1)$\Gamma_\alpha\Gamma_p/\Gamma$=2.6 eV.
9400 *9*	1/2#	2712# *10*	(2J+1)$\Gamma_\alpha\Gamma_p/\Gamma$=0.6 eV.
9456 *4*	3/2#	2776# *4*	(2J+1)$\Gamma_\alpha\Gamma_p/\Gamma$=2.8 eV.
9481.0 *18*	3/2#	2804# *2*	(2J+1)$\Gamma_\alpha\Gamma_p/\Gamma$=18 eV.
9551 *6*	5/2#	2883# *7*	(2J+1)$\Gamma_\alpha\Gamma_p/\Gamma$=1.9 eV.
9673 *6*	1/2#	3021# *7*	(2J+1)$\Gamma_\alpha\Gamma_p/\Gamma$=1.1 eV.
9713.0 *27*	1/2#	3066# *3*	(2J+1)$\Gamma_\alpha\Gamma_p/\Gamma$=41 eV.
9751.1 *27*	7/2#	3109# *3*	(2J+1)$\Gamma_\alpha\Gamma_p/\Gamma$=7.6 eV.
9814.0 *27*	5/2#	3180# *3*	(2J+1)$\Gamma_\alpha\Gamma_p/\Gamma$=20 eV.
9869.8 *27*	1/2	3243 *3*	E_α(lab) (keV): weighted average of 3243 *3* (1964Ku12) and 3242 *5* (1975Sc40). J^{π}: 1/2 from 1964Ku12. (2J+1)$\Gamma_\alpha\Gamma_p/\Gamma$=46 eV.
9900.8 *27*	3/2	3278 *3*	E_α(lab) (keV): weighted average of 3278 *3* (1964Ku12) and 3278 *5* (1975Sc40). J^{π}: 3/2 from 1964Ku12. (2J+1)$\Gamma_\alpha\Gamma_p/\Gamma$=38 eV.
9923 *4*	$3/2^-$	3303 *4*	E_α(lab) (keV): weighted average of 3302 *4* (1964Ku12) and 3305 *5* (1975Sc40). J^{π}: (3/2) from 1964Ku12. (2J+1)$\Gamma_\alpha\Gamma_p/\Gamma$=80 eV.
9958 *6*	3/2	3342 *7*	E_α(lab) (keV): weighted average of 3348 *5* (1971Mc23) and 3335 *5* (1975Sc40). J^{π}: 3/2 from 1971Mc23. (2J+1)$\Gamma_\alpha\Gamma_p/\Gamma$=34 eV.
9968 *5*	(1/2,3/2)@	3354@ *5*	(2J+1)$\Gamma_\alpha\Gamma_p/\Gamma$=17 eV.
10031 *5*	$1/2^+,3/2^-$	3425 *5*	E_α(lab) (keV): weighted average of 3424 *5* (1971Mc23) and 3425 *5* (1975Sc40). J^{π}: (1/2) from 1971Mc23.

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^{31}P(α,p),(α,n):resonances 1975Sc40,1971Mc23,1964Ku12 (continued)

^{35}Cl Levels (continued)

E(level)†	J$^{\pi}$‡	E$_{\alpha}$(lab) (keV)	Comments
			(2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=124 eV.
10058 *5*	3/2@	3455@ *5*	(2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=40 eV.
10075 *5*	(1/2)@	3475@ *5*	(2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=52 eV.
10090 *5*	5/2$^+$	3491 *5*	E$_{\alpha}$(lab) (keV): weighted average of 3491 *5* (1971Mc23) and 3490 *5* (1975Sc40). J$^{\pi}$: 5/2 from 1971Mc23. (2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=63 eV.
10132 *5*	3/2$^-$	3539 *5*	E$_{\alpha}$(lab) (keV): weighted average of 3538 *5* (1971Mc23) and 3540 *5* (1975Sc40). J$^{\pi}$: 3/2 from 1971Mc23. (2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=165 eV.
10172 *5*	3/2$^-$,5/2$^-$	3584 *5*	E$_{\alpha}$(lab) (keV): weighted average of 3587 *5* (1971Mc23) and 3580 *5* (1975Sc40). J$^{\pi}$: (5/2) from 1971Mc23. (2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=26 eV.
10215 *5*	3/2	3633 *5*	E$_{\alpha}$(lab) (keV): weighted average of 3631 *5* (1971Mc23) and 3635 *5* (1975Sc40). J$^{\pi}$: 3/2 from 1971Mc23. (2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=63 eV.
10236 *5*	3/2$^-$	3656 *5*	E$_{\alpha}$(lab) (keV): weighted average of 3656 *5* (1971Mc23) and 3655 *5* (1975Sc40). J$^{\pi}$: 3/2 from 1971Mc23. (2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=271 eV.
10278 *5*	1/2@	3704@ *5*	(2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=15 eV.
10295 *5*	@	3723@ *5*	
10322 *5*	5/2$^+$,7/2$^-$	3753 *5*	E$_{\alpha}$(lab) (keV): weighted average of 3755 *5* (1971Mc23) and 3750 *5* (1975Sc40). J$^{\pi}$: 5/2 from 1971Mc23. (2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=63 eV.
10340 *5*	(3/2,5/2)@	3774@ *5*	(2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=50 eV.
10359 *5*	3/2	3795 *5*	E$_{\alpha}$(lab) (keV): weighted average of 3795 *5* (1971Mc23) and 3795 *5* (1975Sc40). J$^{\pi}$: (3/2) from 1971Mc23. (2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=155 eV.
10385 *5*	@	3825@ *5*	
10399 *5*	3/2$^+$,7/2$^+$	3841 *5*	E$_{\alpha}$(lab) (keV): weighted average of 3842 *5* (1971Mc23) and 3840 *5* (1975Sc40). J$^{\pi}$: (1/2,3/2) from 1971Mc23. (2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=61 eV.
10431 *5*	5/2$^+$	3877 *5*	E$_{\alpha}$(lab) (keV): weighted average of 3876 *5* (1971Mc23) and 3877 *5* (1975Sc40). J$^{\pi}$: 5/2 from 1971Mc23. (2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=66 eV.
10467 *5*	5/2$^+$	3917 *5*	E$_{\alpha}$(lab) (keV): weighted average of 3918 *5* (1971Mc23) and 3915 *5* (1975Sc40). J$^{\pi}$: 5/2 from 1971Mc23. (2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=211 eV.
10486 *5*	5/2$^+$	3939 *5*	E$_{\alpha}$(lab) (keV): weighted average of 3939 *5* (1971Mc23) and 3938 *5* (1975Sc40). J$^{\pi}$: 5/2 from 1971Mc23. (2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=76 eV.
10534 *5*	1/2$^+$,5/2$^+$	3993 *5*	E$_{\alpha}$(lab) (keV): weighted average of 3990 *5* (1971Mc23) and 3995 *5* (1975Sc40). J$^{\pi}$: 1/2 from 1971Mc23. (2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=182 eV.
10587 *5*	3/2$^+$	4053 *5*	E$_{\alpha}$(lab) (keV): weighted average of 4050 *5* (1971Mc23) and 4055 *5* (1975Sc40). J$^{\pi}$: 3/2 from 1971Mc23. (2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=71 eV.
10641 *5*	5/2$^+$	4114 *5*	E$_{\alpha}$(lab) (keV): weighted average of 4113 *5* (1971Mc23) and 4114 *5* (1975Sc40). J$^{\pi}$: 5/2 from 1971Mc23. (2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=272 eV.
10678 *5*	3/2$^+$,5/2$^+$	4156 *5*	E$_{\alpha}$(lab) (keV): weighted average of 4156 *5* (1971Mc23) and 4155 *5* (1975Sc40). J$^{\pi}$: 3/2 from 1971Mc23. (2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=132 eV.
10696 *5*	7/2$^-$	4176 *5*	E$_{\alpha}$(lab) (keV): weighted average of 4174 *5* (1971Mc23) and 4178 *5* (1975Sc40).

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^{31}P(α,p),(α,n):resonances 1975Sc40,1971Mc23,1964Ku12 (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^{π}[‡]	E_{α}(lab) (keV)	Comments
			J^{π}: 7/2 from 1971Mc23. $(2J+1)\Gamma_{\alpha}\Gamma_{p}/\Gamma$=100 eV.
10734 5	$3/2^{+},9/2^{+}$	4219 5	E_{α}(lab) (keV): weighted average of 4218 5 (1971Mc23) and 4220 5 (1975Sc40). J^{π}: 3/2 from 1971Mc23. $(2J+1)\Gamma_{\alpha}\Gamma_{p}/\Gamma$=269 eV.
10760 5	$3/2^{+},7/2^{-}$	4248 5	E_{α}(lab) (keV): weighted average of 4246 5 (1971Mc23) and 4250 5 (1975Sc40).
10800 5	$5/2^{+}$	4293 5	E_{α}(lab) (keV): weighted average of 4290 5 (1971Mc23) and 4295 5 (1975Sc40). J^{π}: 5/2 from 1971Mc23. $(2J+1)\Gamma_{\alpha}\Gamma_{p}/\Gamma$=238 eV.
10816 5	$1/2^{+},3/2^{-}$	4311 5	E_{α}(lab) (keV): weighted average of 4308 5 (1971Mc23) and 4313 5 (1975Sc40). J^{π}: 3/2 from 1971Mc23. $(2J+1)\Gamma_{\alpha}\Gamma_{p}/\Gamma$=345 eV.
10844 5	$3/2^{-}$	4343 5	E_{α}(lab) (keV): weighted average of 4342 5 (1971Mc23) and 4343 5 (1975Sc40). J^{π}: 3/2 from 1971Mc23. $(2J+1)\Gamma_{\alpha}\Gamma_{p}/\Gamma$=923 eV.
10870 5	(5/2,7/2)[@]	4372[@] 5	$(2J+1)\Gamma_{\alpha}\Gamma_{p}/\Gamma$=137 eV.
10886 5	[@]	4390[@] 5	
10909 5	$3/2^{+},5/2^{-}$	4416 5	E_{α}(lab) (keV): weighted average of 4413 5 (1971Mc23) and 4418 5 (1975Sc40).
10928 5	7/2[@]	4438[@] 5	$(2J+1)\Gamma_{\alpha}\Gamma_{p}/\Gamma$=231 eV.
10942 5	[@]	4454[@] 5	
10964 5	$5/2^{+}$	4478 5	E_{α}(lab) (keV): weighted average of 4478 5 (1971Mc23) and 4477 5 (1975Sc40). J^{π}: 5/2 from 1971Mc23. $(2J+1)\Gamma_{\alpha}\Gamma_{p}/\Gamma$=181 eV.
10973 5	[@]	4489[@] 5	
10998 5	$3/2^{+},7/2^{-}$	4517 5	E_{α}(lab) (keV): weighted average of 4517 5 (1971Mc23) and 4517 5 (1975Sc40). J^{π}: (5/2) from 1971Mc23. $(2J+1)\Gamma_{\alpha}\Gamma_{p}/\Gamma$=172 eV.
11034 5	$3/2^{+}$	4557 5	E_{α}(lab) (keV): weighted average of 4556 5 (1971Mc23) and 4558 5 (1975Sc40). J^{π}: 3/2 from 1971Mc23. $(2J+1)\Gamma_{\alpha}\Gamma_{p}/\Gamma$=218 eV.
11063 5	$1/2^{+},9/2^{+}$	4590 5	E_{α}(lab) (keV): weighted average of 4589 5 (1971Mc23) and 4590 5 (1975Sc40).
11082 5	$5/2^{+},9/2^{+}$	4612 5	E_{α}(lab) (keV): weighted average of 4611 5 (1971Mc23) and 4613 5 (1975Sc40).
11105 5	[@]	4638[@] 5	
11123 5	3/2	4658 5	E_{α}(lab) (keV): weighted average of 4657 5 (1971Mc23) and 4658 5 (1975Sc40).
11142 5	$5/2^{+}$	4679 5	E_{α}(lab) (keV): weighted average of 4678 5 (1971Mc23) and 4680 5 (1975Sc40).
11154 5	[@]	4693[@] 5	
11169 5	[@]	4710[@] 5	
11179 5	$3/2^{-},5/2^{+}$	4721 5	E_{α}(lab) (keV): weighted average of 4721 5 (1971Mc23) and 4720 5 (1975Sc40).
11185 5	[@]	4728[@] 5	
11198 5	3/2	4743 5	E_{α}(lab) (keV): weighted average of 4745 5 (1971Mc23) and 4740 5 (1975Sc40).
11230 5	[@]	4779[@] 5	
11246 5	3/2	4797 5	E_{α}(lab) (keV): weighted average of 4799 5 (1971Mc23) and 4795 5 (1975Sc40).
11265 5	$3/2^{-}$	4818 5	E_{α}(lab) (keV): weighted average of 4820 5 (1971Mc23) and 4815 5 (1975Sc40).
11287 5	[@]	4843[@] 5	
11307 5	[@]	4866[@] 5	
11314 5	$1/2^{-},5/2^{+}$	4874 5	E_{α}(lab) (keV): weighted average of 4875 5 (1971Mc23) and 4873 5 (1975Sc40).
11327 5	$1/2^{-},5/2^{+}$	4888 5	E_{α}(lab) (keV): weighted average of 4886 5 (1971Mc23) and 4890 5 (1975Sc40).
11357 5	[@]	4922[@] 5	
11375 5	[@]	4942[@] 5	
11390 5	[@]	4960[@] 5	

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^{31}P(α,p),(α,n):resonances 1975Sc40,1971Mc23,1964Ku12 (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^{π}[‡]	E_{α}(lab) (keV)	Comments
11410 *5*	[@]	4982[@] *5*	
11421 *5*	3/2$^-$	4995 *5*	E_{α}(lab) (keV): weighted average of 4995 *5* (1971Mc23) and 4995 *5* (1975Sc40).
11434 *5*	[@]	5009[@] *5*	
11460 *5*	[@]	5038[@] *5*	
11473 *5*	1/2$^-$,5/2$^-$	5053 *5*	E_{α}(lab) (keV): weighted average of 5056 *5* (1971Mc23) and 5050 *5* (1975Sc40).
11492 *5*	[@]	5075[@] *5*	
11505 *5*	3/2$^-$,5/2$^-$	5089 *5*	E_{α}(lab) (keV): weighted average of 5090 *5* (1971Mc23) and 5088 *5* (1975Sc40).
11525 *5*	[@]	5112[@] *5*	
11540 *5*	[@]	5129[@] *5*	
11551 *5*	3/2$^-$,5/2$^+$	5141 *5*	E_{α}(lab) (keV): weighted average of 5142 *5* (1971Mc23) and 5140 *5* (1975Sc40).
11565 *5*	[@]	5157[@] *5*	
11589 *5*	[@]	5184[@] *5*	
11609 *5*	5/2$^+$,7/2$^-$	5207 *5*	E_{α}(lab) (keV): weighted average of 5209 *5* (1971Mc23) and 5205 *5* (1975Sc40).
11627 *5*	1/2$^+$,5/2$^+$	5227 *5*	E_{α}(lab) (keV): weighted average of 5224 *5* (1971Mc23) and 5230 *5* (1975Sc40).
11638 *5*	[@]	5239[@] *5*	
11651 *5*	[@]	5254[@] *5*	
11684 *5*	1/2$^+$,5/2$^+$	5292[&] *5*	
11773 *5*	3/2$^-$,7/2$^-$	5392[&] *5*	
11783 *5*	3/2$^-$	5403[&] *5*	
12900[a]			T=3/2
13900[a]			T=3/2

† From $E_{c.m.}$+S(α)(^{35}Cl), where S(α)(^{35}Cl)=6997.91 *4* (2021Wa16). $E_{c.m.}$=E_{α}(lab)×m(^{31}P)/[m_{α}+m(^{31}P)].
‡ From analysis of measured angular distributions in 1975Sc40, unless otherwise noted.
# From 1964Ku12, unless otherwise noted.
@ From 1971Mc23, unless otherwise noted.
& From 1975Sc40, unless otherwise noted.
a From 1970Um01. Two ^{35}Cl resonances are found in ^{34}S(p,n) and ^{31}P(α,n), but not in ^{31}P(α,α_0) reactions. T=1 ^{34}S g.s. and T=1/2 protons can populate either T=1/2 or T=3/2 states in ^{35}Cl. T=1/2 ^{31}P g.s. and T=0 α particles can populate T=1/2 states in ^{35}Cl. A small isospin impurity would allow formation of T=3/2 states in ^{35}Cl. The lack of resonances in ^{31}P(α,α_0) indicates that these two resonances are most likely T=3/2. A T=3/2 resonances in ^{31}P(α,α_0) scattering is doubly isospin forbidden (both entrance and exit channels) compared with singly forbidden (entrance channel only) for ^{31}P(α,n).